

7. ANALOG MODULES

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Apart from a large number of digital I/O lines, the PIC16F887 contains 14 analog inputs. They enable the microcontroller to recognize, not only whether a pin is driven to logic zero or one (0 or +5V), but to precisely measure its voltage and convert it into a numerical value, i.e. digital format. The whole procedure takes place in the A/D converter module which has the following features:

- The converter generates a 10-bit binary result using the method of successive approximation and stores the conversion results into the ADC registers (ADRESL and ADRESH);
- There are 14 separate analog inputs;
- The A/D converter allows conversion of an analog input signal to a 10-bit binary representation of that signal; and
- By selecting voltage references Vref- and Vref+, the minimal resolution or quality of conversion may be adjusted to various needs.

ADC Mode and Registers

Even though the use of A/D converter seems to be very complicated, it is basically very simple, simpler than using timers and serial communication module, anyway.

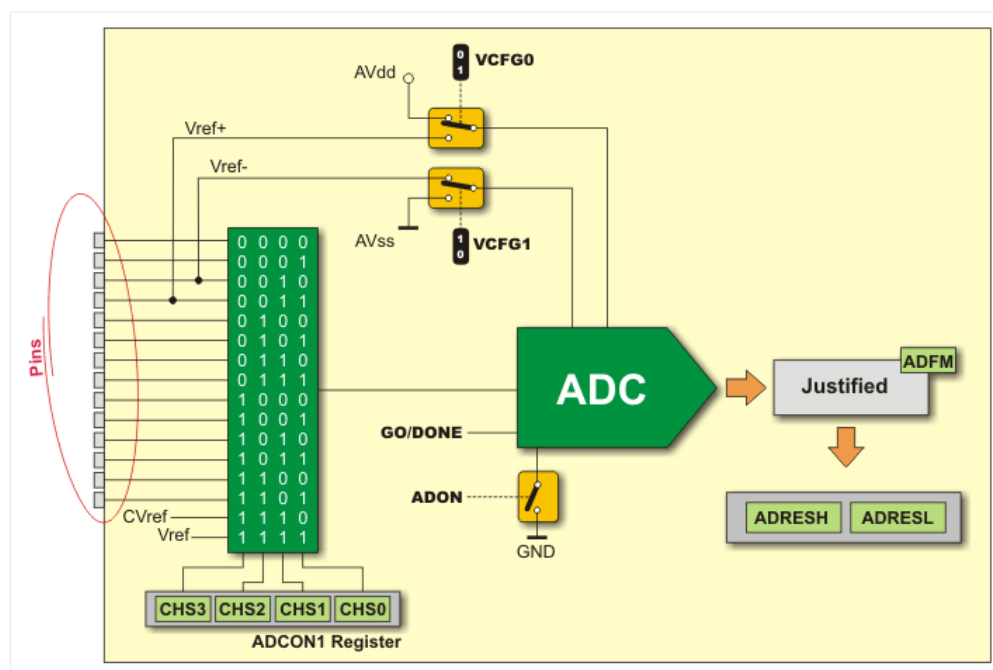


Fig. 7-1 ADC Mode and Registers

The module is under the control of the bits of four registers:

- ADRESH – Contains high byte of conversion result;
- ADRESL – Contains low byte of conversion result;

- ADCON0 – Control register 0; and
- ADCON1 Control register 1

ADRESH and ADRESL Registers

When converting an analog value into a digital one, the result of the 10-bit A/D conversion will be stored in these two registers. In order to deal with this value easier, it can appear in two formats- left justified and right justified. The ADFM bit of the ADCON1 register determines the format of conversion result (see figure 7-2). In the event that A/D converter is not used, these registers may be used as general-purpose registers.

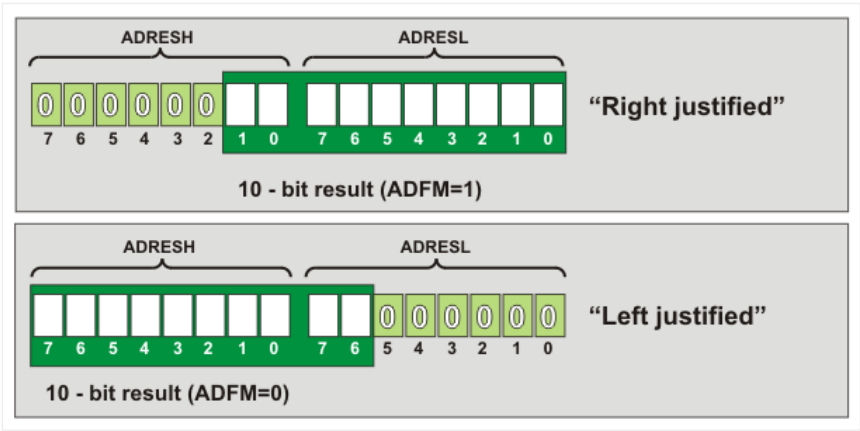


Fig. 7-2 ADRESH and ADRESL Registers

A/D Acquisition Requirements

For the ADC to meet its specified accuracy, it is necessary to provide a certain time delay between selecting specific analog input and measurement itself. This time is called “acquisition time” and mainly depends on the source impedance. There is an equation used for accurately calculating this time, which in the worst case amounts to approximately 20uS. Briefly, after selecting (or changing) the analog input and before starting conversion it is necessary to provide at least 20uS time delay to enable the ACD maximal conversion accuracy.

ADC Clock Period

Time needed to complete a one-bit conversion is defined as TAD. The required TAD must be at least 1,6 uS. One full 10-bit A/D conversion is a bit longer than expected and amounts to 11 TAD periods. However, since both the conversion clock frequency and source are determined by software, one of the available combination of bits ADCS1 and ADCS0 should be selected before voltage measurement on some analog input starts. These bits are stored in the ADCON0 register.

ADC CLOCK SOURCE	ADCS1	ADCS0	DEVICE FREQUENCY (FOSC)			
			20 Mhz	8 Mhz	4 Mhz	1 Mhz
Fosc/2	0	0	100 nS	250 nS	500 nS	2 uS
Fosc/8	0	1	400 nS	1 uS	2 uS	8 uS
Fosc/32	1	0	1.6 uS	4 uS	8 uS	32 uS
Frc	1	1	2 – 6 uS	2 – 6 uS	2 – 6 uS	2 – 6 uS

Table 7-1 ADC Clock Period

Any change in the system clock frequency will affect the ADC clock frequency, which may adversely affect the ADC result. Device frequency characteristics are shown in the table above. The values in the shaded cells are outside of recommended range.

How to Use A/D Converter?

- A/D converter does not differ between digital and analog voltages. In order to avoid errors in measurement or chip damage, the pins should be configured as analog inputs before conversion starts. The bits used for this purpose are stored in the TRIS and ANSELH registers;
- When the port with analog inputs marked as CH0-CH13 is read, the corresponding bits will be driven to logic zero (0); and
- Roughly speaking, voltage measurement in the converter is based on comparing input voltage with internal scale which has 1024 marks ($2^{10}=1024$). The lowest scale mark stands for the V_{ref-} voltage, whilst the highest mark stands for the V_{ref+} voltage. Figure 7-3 below shows selectable referent voltages and their minimum and maximum values as well.

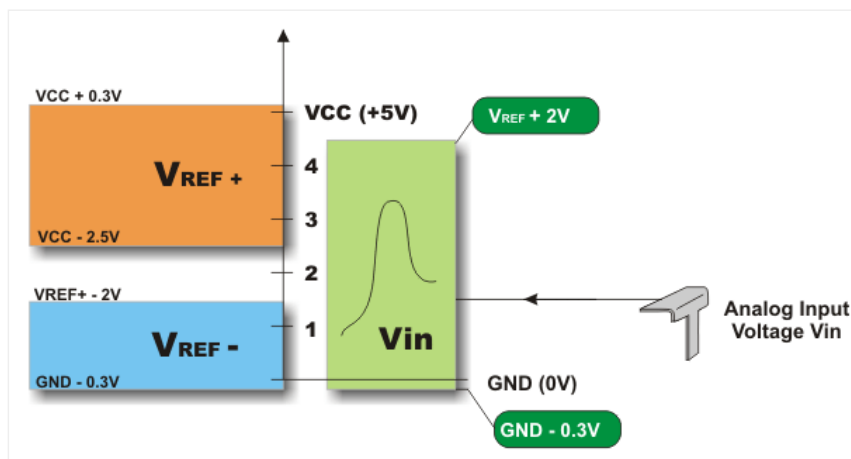


Fig. 7-3 How to Use The A/D Converter

ADCON0 Register

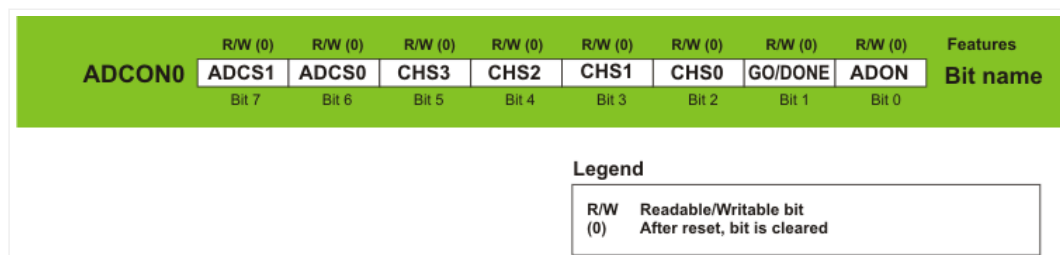


Fig. 7-4 ADCON0 Register

ADCS1, ADCS0 – A/D Conversion Clock Select bits select clock frequency used for internal synchronization of A/D converter. It also affects duration of conversion.

ADCS1	ADCS2	CLOCK
0	0	Fosc/2
0	1	Fosc/8
1	0	Fosc/32
1	1	RC *

Table 7-2 A/D Conversion Select Bits

* Clock is generated by internal oscillator which is built in converter.

CHS3-CHS0 – Analog Channel Select bits select a pin or an analog channel for conversion, i.e. voltage measurement:

CHS3	CHS2	CHS1	CHS0	CHANNEL	PIN
------	------	------	------	---------	-----

0	0	0	0	0	RA0/AN0
0	0	0	1	1	RA1/AN1
0	0	1	0	2	RA2/AN2
0	0	1	1	3	RA3/AN3
0	1	0	0	4	RA5/AN4
0	1	0	1	5	RE0/AN5
0	1	1	0	6	RE1/AN6
0	1	1	1	7	RE2/AN7
1	0	0	0	8	RB2/AN8
1	0	0	1	9	RB3/AN9
1	0	1	0	10	RB1/AN10
1	0	1	1	11	RB4/AN11
1	1	0	0	12	RB0/AN12
1	1	0	1	13	RB5/AN13
1	1	1	0	CVref	
1	1	1	1	Vref = 0.6V	

Table 7-3 Analog Channel Status Bits

GO/DONE – A/D Conversion Status bit determines current status of conversion:

- 1 – A/D conversion is in progress; and
- 0 – A/D conversion is complete. This bit is automatically cleared by hardware when the A/D conversion is completed.

ADON – A/D On bit enables A/D converter.

- 1 – A/D converter is enabled; and
- 0 – A/D converter is disabled.

ADCON1 Register

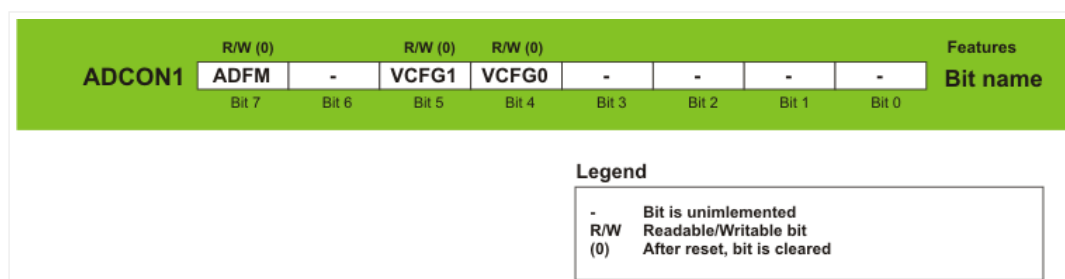


Fig. 7-5 ADCON1 Register

ADFM – A/D Result Format Select bit

- 1 – Conversion result right justified. Six most significant bits of the ADRESLH are not used; and
- 0 – Conversion result left justified. Six least significant bits of the ADRESL are not used.

VCFG1 – Voltage Reference bit selects negative voltage reference source needed for A/D converter operating.

- 1 – Negative voltage reference is applied on the Vref- pin; and
- 0 – Voltage power supply Vss is used as negative voltage reference source.

VCFG0 – Voltage Reference bit selects positive voltage reference source needed for A/D converter operating.

- 1 – Positive voltage reference is applied on the Vref+ pin; and
- 0 – Voltage power supply Vdd is used as positive voltage reference source.

In Short:

In order to measure voltage on an input pin by A/D converter the following should be done:

Step 1 – Configuring port:

- Write logic one (1) to the corresponding bit of the TRIS register to configure it as input; and
- Write logic one (1) to the corresponding bit of the ANSEL register to configure it as analog input.

Step 2 – Configuring ADC module:

- Configure voltage reference in the ADCON1 register;
- Select ADC conversion clock in the ADCON0 register;
- Select one of input channels CH0-CH13 of the ADCON0 register;
- Select data format using the ADFM bit of the ADCON1 register; and
- Enable A/D converter by setting the ADON bit of the ADCON0 register.

Step 3 – Configuring ADC interrupt (optionally):

- Clear the ADIF bit; and
- Set the ADIE, PEIE and GIE bits.

Step 4 – Wait for the required acquisition time (approximately 20uS) to pass.**Step 5** – Start conversion by setting the GO/DONE bit of the ADCON0 register.**Step 6** – Wait for ADC conversion to complete.

- It is necessary to check in program loop whether the GO/DONE pin is cleared or wait for an A/D interrupt (must be previously enabled).

Step 7 – Read ADC results:

- Read the ADRESH and ADRESL registers.

Analog Comparator

In addition to A/D converter, there is one more module, which until quite recently has been embedded only in integrated circuits, belonging to so called analog electronics. Owing to the fact that it is hardly possible to find any more complex automatic device which in some way does not use these circuits, two high quality comparators along with additional electronics are integrated into the microcontroller and connected to its pins.

How does a comparator operate? Basically, the analog comparator is an amplifier which compares the magnitude of voltages at two inputs. Looking at its physical features, it has two inputs and one output. Depending on which input has a higher voltage (analog value), a logic zero (0) or logic one (1) (digital values) will appear on its output:

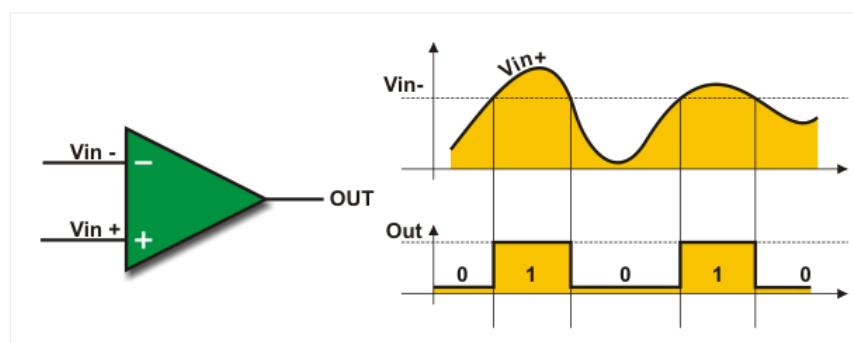


Fig. 7-6 Analog Comparator

- The PIC16F887 microcontroller has two such voltage comparators whose inputs are connected to I/O pins RA0-RA3, whereas the outputs are connected to the pins RA4 and RA5. In addition there is also a referent voltage internal source on chip itself, but it will be discussed later.

CM1CON0 is in control of comparator C1;
CM2CON0 is in control of comparator C2; and
CM2CON1 is in control of comparator C2.

One of two analog voltages provided on the comparator inputs is usually stable and unchangeable. Because of those features it is called “voltage reference”(Vref). To generate it, both external and special internal voltage source can be used. After selecting voltage source, Vref is derived from it by means of ladder network consisting of 16 resistors which form voltage divider. The voltage source is selectable through both ends of that divider through the VRSS bit of the VRCON register.

The comparator voltage reference has 2 ranges with 16 voltage levels in each range. Range selection is controlled by the VRR bit of the VRCON register. The selected voltage reference may be output to the RA2/AN2 pin.

It's operation is under control of the VRCON register.

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The flag bit CMIF of the register PIR is set on every change of logic state on any comparator's output. The same changes also cause an interrupt if the following bits are set:

CMIE bit of the PIE register;
PEIE bit of the INTCON register; and
GIE bit of the INTCON register.

If interrupt is enabled, any change on the comparator's output can wake up the microcontroller from *sleep mode* if it is setup in that mode.

CM1CON0 Register

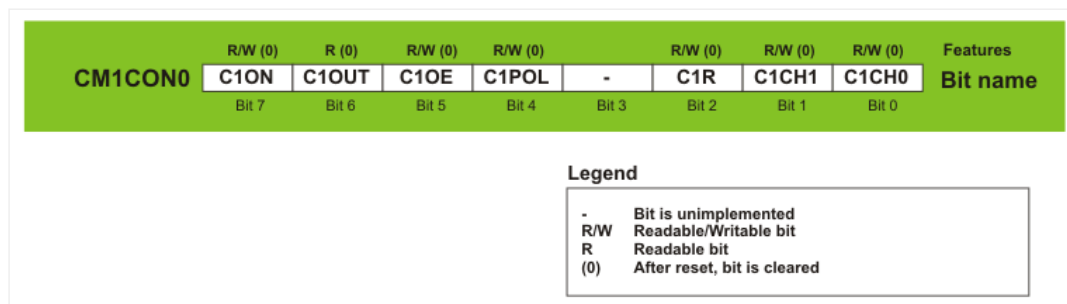


Fig. 7-8 CM1CON0 Register

Bits of this register are in control of the comparator C1. It mainly affects configuration of its inputs. To understand it better, look at figure 7-9 below which shows only a part of electronics directly affected by the bits of this register.

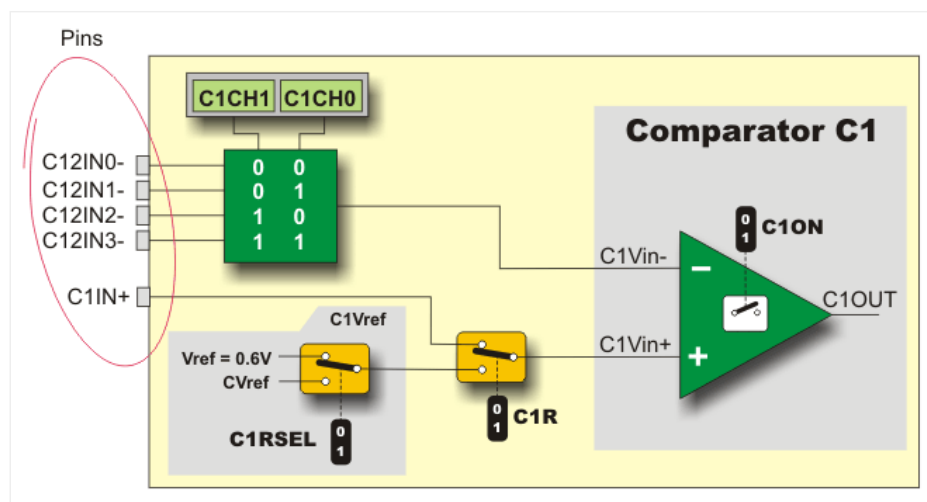


Fig. 7-9 Comparator C1 Enable Bit

C1ON – Comparator C1 Enable bit enables comparator C1.

- 1 – Comparator C1 is enabled; and
- 0 – Comparator C1 is disabled.

C1OUT – Comparator C1 Output bit is comparator C1 output bit.

If C1POL = 1 (comparator output is inverted)

- 1 – Analog voltage at C1Vin+ is lower than analog voltage at C1Vin-; and
- 0 – Analog voltage at C1Vin+ is higher than analog voltage at C1Vin-.

If C1POL = 0 (comparator output is non-inverted)

- 1 – Analog voltage at C1Vin+ is higher than analog voltage at C1Vin-; and
- 0 – Analog voltage at C1Vin+ is lower than analog voltage at C1Vin-.

C1OE Comparator C1 Output Enable bit.

- 1 – Comparator C1OUT output is connected to the C1OUT pin.*; and
- 0 – Comparator output is internal only.

* In order to enable the C1OUT bit to be present on the pin, two conditions must be met: C1ON = 1 (comparator must be on) and the corresponding TRIS bit = 0 (pin must be configured as output).

C1POL – Comparator C1 Output Polarity Select bit enables comparator C1 out put state to be inverted.

- 1 – Comparator C1 output is inverted; and
- 0 – Comparator C1 output is non-inverted.

C1R – Comparator C1 Reference Select bit

- 1 – Non-inverting input C1Vin+ is connected to reference voltage C1Vref; and
- 0 – Non-inverting input C1Vin+ is connected to the C1IN+ pin.

C1CH1, C1CH0 – Comparator C1 Channel Select bit

C1CH1	C1CH0	COMPARATOR C1VIN- INPUT
0	0	Input C1Vin- is connected to the C12IN0- pin
0	1	Input C1Vin- is connected to the C12IN1- pin
1	0	Input C1Vin- is connected to the C12IN2- pin
1	1	Input C1Vin- is connected to the C12IN3- pin

Table 7-4 Comparator C1

CM2CON0 Register

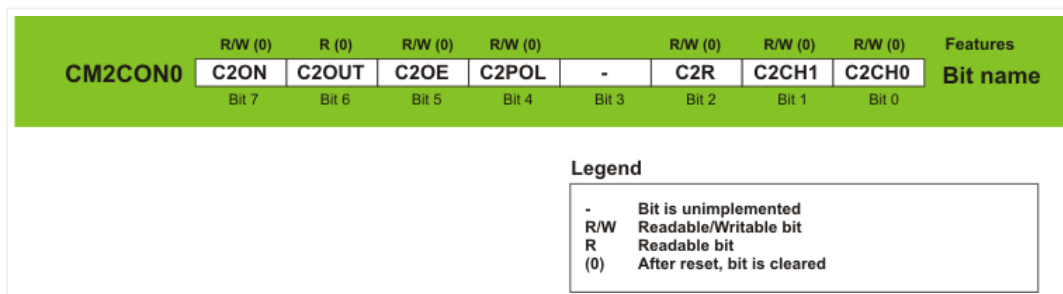


Fig. 7-10 CM2CON0 Register

Bits of this register are in control of comparator C2. Similar to the previous case, the figure 7-11 shows a simplified schematic of the circuit affected by the bits of this register.

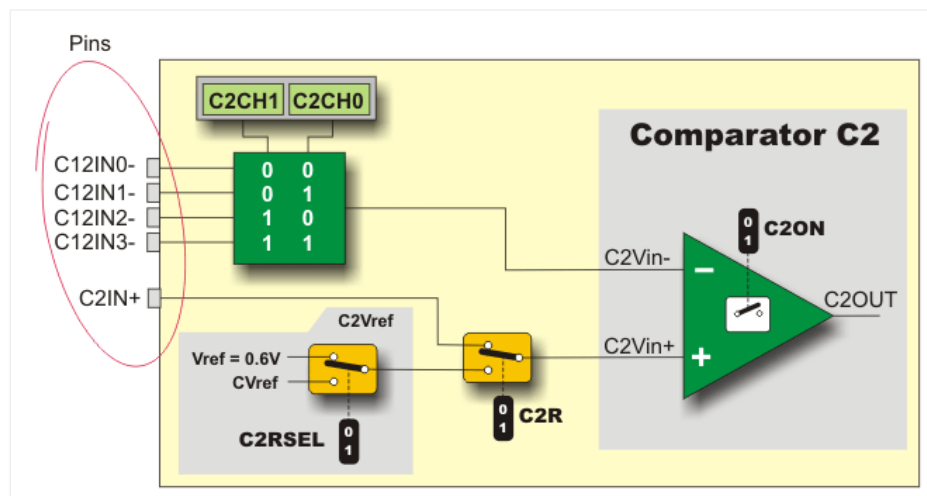


Fig. 7-11 Comparator C2 Schematic Diagram

C2ON – Comparator C2 Enable bit enables comparator C2.

- 1 – Comparator C2 is enabled; and
- 0 – Comparator C2 is disabled.

C2OUT – Comparator C2 Output bit is comparator C2 output.

If C2POL = 1 (comparator output inverted)

- 1 – Analog voltage at C1Vin+ is lower than analog voltage at C1Vin-; and
- 0 – Analog voltage at C1Vin+ is higher than analog voltage at C1Vin-.

If C2POL = 0 (comparator output non-inverted)

- 1 – Analog voltage at C1Vin+ is higher than analog voltage at C1Vin-; and
- 0 – Analog voltage at C1Vin+ is lower than analog voltage at C1Vin-.

C2OE – Comparator C2Output Enable bit

- 1 – Comparator C2OUT output is connected to the C2OUT pin.*; and
- 0 – Comparator output is internal only.

* In order to enable the C2OUT bit to be present on the pin, two conditions must be met: C2ON = 1 (comparator must be on) and the corresponding TRIS bit = 0 (pin must be configured as output).

C2POL – Comparator C2 Output Polarity Select bit enables comparator C2 out put state to be inverted.

- 1 – Comparator C2 output is inverted; and
- 0 – Comparator C2 output is non-inverted.

C2R – Comparator C2 Reference Select bit

- 1 – Non-inverting input C2Vin+ is connected to reference voltage C2Vref; and
- 0 – Non-inverting input C2Vin+ is connected to the C2IN+ pin.

C2CH1, C2CH0 Comparator C2 Channel Select bit

C2CH1	C2CH0	COMPARATOR C2VIN- INPUT
0	0	Input C2Vin- is connected to the C12IN0- pin
0	1	Input C2Vin- is connected to the C12IN1- pin
1	0	Input C2Vin- is connected to the C12IN2- pin
1	1	Input C2Vin- is connected to the C12IN3- pin

Table 7-5 Comparator C2 Channel Select Bit

CM2CON1 Register

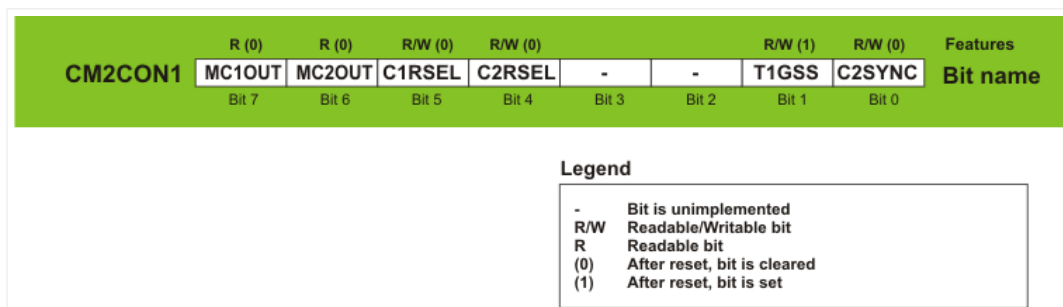


Fig. 7-12 CM2CON1 Register

MC1OUT Mirror Copy of C1OUT bit

MC2OUT Mirror Copy of C2OUT bit

C1RSEL Comparator C1 Reference Select bit

- 1 – Selectable voltage CVref is used in voltage reference C1Vref source; and
- 0 – Fixed voltage reference 0.6V is used in voltage reference C1Vref source.

C2RSEL – Comparator C2 Reference Select bit

- 1 – Selectable voltage CVref is used in voltage reference C2Vref source; and
- 0 – Fixed voltage reference 0.6V is used in voltage reference C2Vref source.

T1GSS – Timer1 Gate Source Select bit

- 1 – Timer T1gate source is T1G; and
- 0 – Timer T1gate source is comparator SYNCC2OUT.

C2SYNC – Comparator C2 Output Synchronization bit

- 1 – Comparator C2 output is synchronized to falling edge of Timer TMR1 clock; and
- 0 – Comparator output is asynchronous signal.

VRCON Register

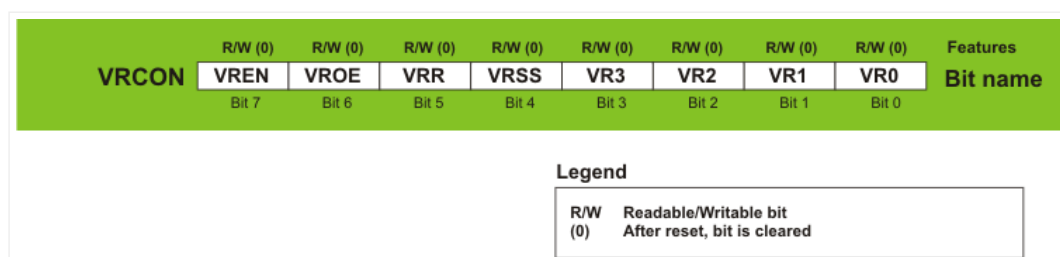


Fig. 7-13 VRCON Register

VREN Comparator C1 Voltage Reference Enable bit

- 1 – Voltage reference CVref source is powered on; and
- 0 – Voltage reference CVref source is powered off.

VROE Comparator C2 Voltage Reference Enable bit

- 1 – Voltage reference CVref is connected to the pin; and
- 0 – Voltage reference CVref is disconnected from the pin.

VRR – CVref Range Selection bit

- 1 – Voltage reference source is set to low range; and
- 0 – Voltage reference source is set to high range.

VRSS – Comparator Vref Range selection bit

- 1 – Voltage reference source is in the range of Vref+ to Vref-; and
- 0 – Voltage reference source is in the range of Vdd – Vss (power supply voltage).

VR3 – VR0 CVref Value Selection

If VRR = 1 (low range)

Voltage reference is calculated using the formula: $CVref = ([VR3:VR0]/24)Vdd$

If VRR = 0 (high range)

Voltage reference is calculated using the formula: $CVref = Vdd/4 + ([VR3:VR0]/32)Vdd$

In Short:

In order to properly use built in Comparators, it is necessary to do the following:

Step 1 – Configuring module:

- In order to select the appropriate mode, bits of the registers CM1CON0 and CM2CON0 should be configured. Interrupt should be disabled on any change of mode.

Step 2 – Configuring internal voltage reference Vref source (only when used). In the VRCON register it is necessary to :

- Select one of two voltage ranges using the VRR bit;
- Configure necessary Vref using bits VR3 – VR0;
- Set the VROE bit if needed; and
- Enable voltage Vref source by setting the VREN bit.

Formula used to calculate voltage reference:

VRR = 1 (low range)

$$CV_{ref} = ([VR3:VR0]/24)VLADDER$$

VRR = 0 (high range)

$$CV_{ref} = (VLADDER/4) + ([VR3:VR0]VLADDER/32)$$

$$V_{ladder} = V_{dd} \text{ or } ([V_{ref+}] - [V_{ref-}]) \text{ or } V_{ref+}$$

Step 3 – Starting operation:

- Enable interrupt by setting bits CMIE (PIE register), PEIE and GIE (both in the INTCON register);
- Read bits C1OUT and C2OUT of the CMCON register; and
- Read flag bit CMIF of the PIR register. After being set, this bit must be cleared in software.



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